# SK Hynix

## Investment Analysis

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*This analysis is based on training data through mid-2025. Figures and leadership details should be verified against the most recent filings.*

# 1. Business History

SK Hynix traces its origins to February 1983, when Hyundai Electronics was established as a semiconductor subsidiary of the Hyundai Group, South Korea's then-dominant chaebol. Founded under the direction of Hyundai patriarch Chung Ju-yung, the venture reflected the Korean government's industrial policy push to build a domestic semiconductor industry capable of competing with Japan. Hyundai Electronics began by manufacturing 64K DRAM chips under technology licenses from Texas Instruments and, within a decade, had clawed its way into the ranks of the world's top DRAM producers through aggressive capital expenditure and a willingness to endure cyclical losses that would have broken Western competitors.

The Asian financial crisis of 1997 forced a brutal restructuring of the Korean chaebol system. Hyundai Group splintered, and in 2001 Hyundai Electronics merged with LG Semiconductor — itself a castoff from LG Group's own restructuring — to form Hynix Semiconductor. The merger created the world's second-largest DRAM maker but saddled the new entity with crushing debt. Hynix survived only through a controversial creditor-led bailout orchestrated by Korea Exchange Bank and other state-linked lenders, an episode that drew anti-subsidy complaints from Micron and triggered countervailing duties in the United States and European Union. This near-death experience permanently embedded a culture of cost discipline and cyclical paranoia into the organization.

The second decisive inflection came in February 2012, when SK Telecom, the flagship of SK Group, acquired a 21.1% controlling stake in Hynix for approximately 3.4 trillion won ($3.1 billion). The company was rebranded SK Hynix in March 2012. SK Group's stewardship brought financial stability, governance reform, and a longer-term investment horizon. Under CEO Park Sung-hwan and later Park Jung-ho, SK Hynix diversified beyond commodity DRAM into higher-margin specialty memory, most critically High Bandwidth Memory (HBM) for AI accelerators. The company shipped the world's first HBM product in 2013 and, by 2024, held a commanding lead in HBM3E supply to NVIDIA, transforming itself from a cyclical commodity player into a strategic enabler of the AI infrastructure buildout.

SK Hynix further reshaped its portfolio through the $9 billion acquisition of Intel's NAND flash and SSD business, announced in October 2020 and completed in two stages — the Dalian fab in December 2021 and the remaining SSD assets finalized by 2025. This deal created Solidigm, a wholly owned NAND subsidiary, and elevated SK Hynix to the number two position in NAND flash globally behind Samsung. The cultural thread connecting the 1983 startup to today's AI-memory champion is a consistent bet-the-company posture toward capital investment through downturns, a pattern inherited from Hyundai's chaebol-era ethos and refined by the near-bankruptcy of the early 2000s into a more calculated but still aggressive approach to capacity and technology leadership.

## 2. Management & Incentives

SK Hynix is led by CEO Kwak Noh-Jung, who assumed the role in March 2024 after serving as president and head of the company's DRAM development division. Kwak is a career semiconductor engineer who joined Hyundai Electronics (SK Hynix's predecessor) in the 1990s and played a central role in developing the company's high-bandwidth memory (HBM) products, which have become the dominant revenue growth driver amid the AI infrastructure buildout. His appointment signaled continuity and a deepening commitment to advanced memory technology. Prior CEO Lee Seok-Hee, who led the company from 2018 through early 2024, oversaw a significant strategic pivot toward premium memory products and navigated the severe cyclical downturn of 2022–2023 before handing the reins to Kwak. CFO Kim Woo-Hyun has provided financial stewardship through the cycle, maintaining discipline on capital expenditure timing despite intense competitive pressure to expand capacity.

SK Hynix is 20.07% owned by SK Square, the investment holding arm of SK Group, South Korea's second-largest chaebol controlled by Chairman Chey Tae-Won. SK Group's effective economic interest, combined with cross-holdings through SK Telecom and SK Square, gives the parent group decisive influence over strategic direction, board appointments, and capital allocation. This concentrated ownership structure provides stability and long-term strategic alignment but raises familiar Korean governance concerns around minority shareholder protection and related-party transactions. Notably, SK Group has historically treated SK Hynix as a crown jewel asset, channeling significant group resources toward its competitiveness, including the $9 billion acquisition of Intel's NAND business (Solidigm) completed in 2021.

Compensation at SK Hynix follows the Korean conglomerate norm: base salaries for senior executives are modest by global semiconductor standards, with significant performance bonuses tied to operating profit targets and strategic milestones. Long-term equity-based incentive plans remain less prominent than at Western peers like Micron, which somewhat limits direct alignment between executive wealth and share price performance. That said, SK Group's concentrated ownership means the controlling shareholder's economic interests are deeply tied to the stock.

Capital allocation has been disciplined relative to peers. Management resisted the temptation to aggressively expand commodity DRAM and NAND capacity during the 2021 upcycle, instead prioritizing the HBM product roadmap. This proved prescient: SK Hynix secured an estimated 50%+ share of the HBM3E market by 2024, becoming NVIDIA's primary supplier. The Solidigm acquisition, initially questioned for its timing, has shown improving execution. Dividend policy has been conservative, with management prioritizing balance sheet repair after the 2023 trough.

The board includes independent directors as required under Korean governance reforms, though true independence from SK Group influence is debatable, a structural feature common across chaebol-affiliated companies. Succession planning is managed within the SK Group framework, reducing key-person risk at any single executive level but concentrating ultimate decision authority in the group chairman. Kwak's elevation from within the technical ranks suggests a functioning internal pipeline, though the company's fortunes remain heavily tied to SK Group's broader strategic priorities. On credibility, management has largely delivered on stated targets around HBM leadership and technology node transitions, building a stronger execution track record over the past three years than the prior decade would have suggested.

# 3. Competitive Landscape

The global memory semiconductor industry operates as a tight oligopoly. In DRAM, three firms — Samsung Electronics, SK Hynix, and Micron Technology — collectively control approximately 95% of revenue, with Samsung historically leading at roughly 40–45%, SK Hynix holding around 30–35%, and Micron capturing 20–25%. In NAND flash, the field is somewhat broader, adding Kioxia (formerly Toshiba Memory), Western Digital, and Solidigm (an SK-affiliated entity through Intel's NAND acquisition), but the top three DRAM players still dominate. This concentrated structure, the product of decades of brutal capital-intensity-driven consolidation, is itself one of the industry's most important competitive features: it enables periods of rational supply discipline that were impossible when a dozen producers competed.

SK Hynix's most consequential competitive advantage today is its leadership in High Bandwidth Memory (HBM), the specialized DRAM stacked using through-silicon vias that is essential for AI accelerators. SK Hynix was the first to mass-produce HBM3E and has been NVIDIA's preferred supplier, reportedly capturing over 50% of the HBM market and securing the vast majority of NVIDIA's HBM allocation through advance contracts. This position reflects years of process engineering investment and close co-development with NVIDIA — a relationship that functions as a significant switching cost for both parties. Samsung has struggled with yield issues on its competing HBM3E products, and Micron, while qualifying with NVIDIA, has lagged in volume. HBM commands margins substantially above commodity DRAM, and SK Hynix's first-mover advantage here has transformed its earnings profile.

Barriers to entry in memory are formidable. A greenfield DRAM fab costs $15–20 billion and requires years of process know-how that cannot simply be purchased. China's CXMT represents the only credible new entrant in DRAM, but it remains several technology generations behind and faces ongoing U.S. export controls that restrict its access to extreme ultraviolet (EUV) lithography equipment. In NAND, China's YMTC has made faster progress but likewise operates under export restrictions that cap its competitiveness at the frontier.

The durability of SK Hynix's advantages is moderate to high but not impregnable. In commodity DRAM and NAND, pricing power is cyclical; memory remains a largely fungible product where customers qualify multiple suppliers and shift orders based on price and availability. SK Hynix's HBM moat is more defensible in the near term given the tight engineering integration required, but Samsung is investing aggressively to close the gap, and any broadening of the AI accelerator vendor base beyond NVIDIA could dilute SK Hynix's privileged position. In the value chain, SK Hynix sits as a critical upstream supplier to hyperscalers, OEMs, and AI chip designers — a position of structural importance but one that also exposes it to the concentrated bargaining power of a small number of enormous customers. The most disruptive longer-term threats include emerging non-volatile memory technologies such as compute-in-memory architectures that could reduce DRAM demand intensity, and the potential for advanced packaging innovations to shift value away from memory producers and toward foundries or system integrators. For now, however, the combination of oligopoly structure, capital barriers, process complexity, and AI-driven HBM demand provides SK Hynix with a competitively advantaged position that is likely to persist through the current investment cycle.

# 4. Financial Characteristics

SK Hynix derives virtually all of its revenue from semiconductor memory, with DRAM historically constituting roughly 70–75% of sales and NAND flash comprising the remainder. Within DRAM, the company has become the dominant supplier of High Bandwidth Memory (HBM), a product category that has reshaped its revenue mix and margin profile since 2024. HBM — used in AI accelerators from NVIDIA and others — commands substantial price premiums over conventional DRAM and has grown to represent an estimated 30%+ of DRAM revenue. Geographically, revenue skews heavily toward Asia-Pacific, with China, the United States, and other markets housing major hyperscaler and OEM customers.

The company's margin structure is highly cyclical, reflecting memory industry supply-demand dynamics. After a severe downturn in 2023 — when gross margins collapsed to roughly negative 10% and operating margins fell to approximately negative 20% amid an industry-wide inventory correction — SK Hynix staged a dramatic recovery in 2024. Full-year 2024 revenue approximately doubled year-over-year to around KRW 66 trillion (roughly $48 billion), with gross margins recovering to the mid-40s percent range and operating margins reaching approximately 30%. This recovery was powered by both a conventional memory price upcycle and structural demand from AI-related HBM. By Q4 2024, quarterly operating margins were approaching 40%, among the highest in the company's history. Net margins followed a similar trajectory, swinging from deeply negative in 2023 to the mid-20s percent range for full-year 2024.

Returns on capital reflect the same cyclicality. ROE swung from roughly negative 15% in 2023 to over 40% in 2024, while ROIC followed a comparable pattern. At peak cycle, SK Hynix generates returns well above its cost of capital; at trough, it destroys value — a pattern common to memory peers Samsung and Micron, though SK Hynix's HBM leadership has introduced a degree of structural differentiation that may dampen future trough severity.

Cash conversion is strong at cycle peaks. Operating cash flow in 2024 was approximately KRW 28–30 trillion, comfortably funding elevated capital expenditure of roughly KRW 15–18 trillion, yielding meaningful positive free cash flow. Capital intensity is inherently high in memory fabrication: maintenance capex alone runs in the range of KRW 8–10 trillion annually, with growth capex — particularly for HBM advanced packaging and next-generation DRAM process nodes — adding substantially on top. The company has guided capex to rise further as it expands HBM production capacity.

The balance sheet improved markedly through 2024. Net debt declined from a peak of over KRW 20 trillion during the 2023 downturn to a more moderate level as free cash flow accumulated, with the net debt-to-EBITDA ratio dropping below 0.5x. This compares favorably to the leveraged position typical of trough periods. Working capital dynamics are characterized by relatively short inventory cycles (typically 50–70 days), though inventory days spiked above 100 during the 2023 oversupply period before normalizing.

What distinguishes SK Hynix's financial profile from peers is the HBM franchise. The company holds an estimated 50%+ share of the HBM market, a position that generates structurally higher margins than commodity DRAM, provides greater pricing power, and supports more predictable revenue through longer-term supply agreements with hyperscalers. This differentiates it from Samsung (which is broader but has lagged in HBM yield) and Micron (which is catching up but remains subscale in HBM). The risk is that this advantage narrows as competitors close the technology gap, and the underlying business remains exposed to the pronounced boom-bust cyclicality endemic to memory semiconductors, where peak-to-trough revenue swings of 40–50% are not uncommon.

# 5. Red Flags

SK Hynix's position as the dominant HBM supplier to Nvidia is simultaneously its greatest asset and its most dangerous vulnerability. An estimated 70–80% of Nvidia's HBM procurement has flowed through SK Hynix, creating a customer concentration problem that the market has largely chosen to ignore. If Nvidia diversifies its HBM sourcing toward Samsung or Micron — and it has every incentive to do so — SK Hynix's pricing power erodes rapidly. Worse, if AI capex spending decelerates or a major hyperscaler pauses orders, SK Hynix sits on expensive advanced-packaging capacity with limited alternative demand. The company has been here before: the memory industry's history is a graveyard of cycle-peak investment decisions, and SK Hynix itself nearly went bankrupt in 2012 before SK Group's rescue.

The $9 billion acquisition of Intel's NAND business (Solidigm), completed in staged payments through 2025, deserves forensic scrutiny. SK Hynix took on substantial debt to fund a deal for a structurally weaker NAND operation at a time when NAND margins were deeply negative. The integration has consumed management attention and capital while contributing little to earnings. Investors should ask whether the carrying value of Solidigm's assets has been impaired sufficiently, or whether management is amortizing goodwill and intangibles on an optimistic schedule to avoid headline write-downs. The staggered payment structure also obscured the true cost of capital deployed.

Governance at SK Hynix cannot be evaluated in isolation from SK Group, one of South Korea's most complex chaebol structures. Related-party transactions with SK Group affiliates — including SK Telecom, SK Square (the intermediate holding company), and SK Materials — create opacity around transfer pricing, shared services, and capital allocation. SK Group's cross-shareholding web means that minority shareholders in SK Hynix are effectively subordinate to group-level strategic priorities. The 2023 restructuring that placed SK Hynix under SK Square rather than directly under SK Telecom added another layer of holding-company complexity without improving transparency. Management entrenchment is a structural feature of the chaebol model, and outside investors have limited mechanisms to influence board composition or capital return policy.

On the balance sheet, SK Hynix's capital expenditure cycle is aggressive. The company has guided for over 10 trillion won in annual capex to build out HBM and advanced DRAM capacity, funded partly by the cyclical upswing's cash flows but also by rising debt. Net debt-to-EBITDA looks comfortable at the top of the cycle, but memory EBITDA can collapse 80% or more in a downturn — as it did in 2019 and 2023 — transforming a manageable leverage ratio into a crisis-level figure within two quarters. Cash conversion also warrants attention: inventory build-ups of advanced HBM wafers and work-in-progress tie up working capital, and if demand slips, those inventories become impairment candidates rather than assets.

Historically, every memory upcycle has ended with overinvestment. SK Hynix, Samsung, and Micron collectively announce capacity expansions at cycle peaks, and the resulting supply glut craters pricing. The current AI-driven HBM cycle may prove different in duration, but the structural incentive to overproduce remains identical. Regulators also loom: U.S. export controls on advanced semiconductor equipment to China directly threaten SK Hynix's Dalian DRAM fab, which has operated under fragile exemptions. Loss of that exemption would strand billions in invested capital.

The bear case is straightforward: AI spending normalizes, HBM demand plateaus, Samsung closes the technology gap and ignites a price war, NAND remains structurally oversupplied, and SK Hynix is left with a bloated cost structure, an underperforming Solidigm subsidiary, and a leveraged balance sheet entering a downturn. Investors should be pressing management on three questions: What is the contractual visibility on HBM orders beyond the next two quarters? What triggers would prompt a capex reduction? And what is the realistic standalone valuation of Solidigm today versus its book value?

# 6. Investment Checklist

1. **Understandable Business: YES** SK Hynix manufactures memory semiconductors — primarily DRAM and NAND flash — which serve as essential storage and processing components in smartphones, servers, PCs, and AI accelerators. The business model is straightforward: fabricate silicon wafers into memory chips and sell them to OEMs and hyperscalers. Revenue is driven by bit shipment volumes and average selling prices, both of which are transparent and trackable.

2. **Durable Competitive Advantages (Widening Not Narrowing): YES** The global memory industry has consolidated into a tight oligopoly — Samsung, SK Hynix, and Micron control virtually all DRAM supply (approximately 95%+). SK Hynix holds roughly 35% DRAM market share and has established a commanding lead in High Bandwidth Memory (HBM), the critical memory for AI accelerators. It was the first to mass-produce HBM3E and has been NVIDIA's preferred HBM supplier. The capital intensity required to enter or compete at the leading edge creates formidable barriers. The HBM lead, built on advanced packaging and process expertise, appears to be widening as next-generation HBM4 development proceeds.

3. **Long Reinvestment Runway at High Returns: PARTIALLY** The AI-driven demand for HBM and high-performance memory provides a multi-year growth vector, with HBM TAM expected to grow from roughly $16 billion in 2024 to potentially $60–100 billion by 2028. However, memory remains a cyclical industry where reinvestment returns fluctuate with supply-demand dynamics. During upcycles, returns on capital are exceptional; during downturns, they can turn negative. The HBM opportunity extends the runway for high-return reinvestment, but the structural cyclicality of commodity memory tempers confidence in sustained high returns across the full cycle.

4. **High Returns on Incremental Capital: PARTIALLY** In the current AI-driven upcycle, returns on incremental capital have been extraordinary. SK Hynix reported record operating profit of approximately KRW 23.5 trillion in 2024, with operating margins exceeding 40% in peak quarters. HBM commands substantially higher margins than commodity DRAM. However, through-cycle ROIC has historically averaged in the mid-teens at best, with significant variance. The incremental capital going into HBM and advanced packaging likely earns well above the cost of capital, but conventional DRAM and NAND investments deliver more moderate and cyclical returns.

5. **Strong Free Cash Flow Generation: PARTIALLY** SK Hynix generated significant operating cash flow during the 2024 upcycle, but free cash flow is constrained by heavy capital expenditure requirements. Capital intensity typically runs 30–40% of revenue. Free cash flow is strongly positive during upcycles but can turn negative during downturns when revenue falls but capex commitments persist. The structural need for continuous heavy investment limits cumulative free cash flow conversion relative to reported earnings.

6. **Conservative Balance Sheet: PARTIALLY** SK Hynix substantially improved its balance sheet through the 2024 upcycle after taking on significant debt during the 2023 downturn and the Solidigm acquisition. Net debt was reduced meaningfully through 2024. However, the company still carries meaningful gross debt, and the cyclical nature of the business means leverage ratios can deteriorate quickly during downturns. The balance sheet is adequate but not fortress-like; it is managed pragmatically rather than conservatively.

7. **Aligned Management with Skin in the Game: PARTIALLY** SK Hynix is controlled by SK Group (SK Square holds approximately 20%). CEO Kwak Noh-Jung has demonstrated strong execution on HBM strategy. However, the chaebol governance structure raises typical Korean corporate governance concerns — minority shareholder interests may not always be paramount, and cross-holdings within the SK Group ecosystem create potential conflicts. Direct insider ownership by

management is relatively limited compared to founder-led Western companies.

**8. Rational Capital Allocation History:** **PARTIALLY** Recent capital allocation has been impressive — the early and aggressive bet on HBM has proven visionary. The Solidigm acquisition was strategically motivated but came at a challenging point in the NAND cycle. Dividend policy has been modest by global standards. Capital allocation is improving but has historically reflected chaebol priorities over pure shareholder return maximization.

**9. Favourable Industry Structure:** **YES** Three players control the DRAM market; four to five control NAND. This oligopoly structure has led to improved supply discipline. HBM further improves the structure because it requires advanced packaging capabilities that even fewer competitors possess — effectively a duopoly between SK Hynix and Samsung for cutting-edge HBM. Pricing power in HBM is meaningfully stronger than in commodity memory.

**10. Reasonable Valuation Relative to Quality and Growth: UNCLEAR.** Without access to the current share price and real-time consensus estimates as of March 2026, a precise valuation assessment is not possible. Investors should be cautious about paying peak-cycle multiples on peak-cycle earnings, a pattern that has historically punished memory investors. The quality of the HBM franchise may justify a structural re-rating, but cyclical risk remains embedded in the earnings stream.

**11. Resilience to Recession or Disruption:** **NO** Memory semiconductors are among the most cyclical subsectors in technology. SK Hynix's revenue declined roughly 30% in 2023 and it posted operating losses during that downturn. A recession that curtails enterprise IT spending, smartphone demand, or AI infrastructure investment would hit results hard. The high fixed-cost manufacturing base amplifies earnings volatility in both directions.

**12. Transparent Management Communication:** **PARTIALLY** SK Hynix holds quarterly earnings calls and provides guidance on market conditions, technology roadmaps, and demand trends. Disclosure quality has improved in recent years. However, the company does not provide granular segment-level margins (e.g., HBM-specific profitability), and communication tends to be less detailed than US-listed peers like Micron.

## Overall Assessment

SK Hynix is an exceptionally well-positioned company within a structurally attractive oligopoly, riding the most significant demand wave in memory history through its HBM leadership. The business is understandable, the competitive moat is widening in the near term, and industry structure is favourable. However, investors must weigh these strengths against the inherent deep cyclicality of memory semiconductors, heavy capital intensity that constrains free cash flow, chaebol governance risks, and limited resilience to macroeconomic or AI spending downturns. SK Hynix is best understood as a high-quality cyclical with a powerful secular growth overlay — not a compounder. Position sizing and entry valuation discipline are critical, as the difference between buying at the right and wrong point in the memory cycle has historically been the dominant driver of investor returns.

# 7. Synthesis

## The 3–4 Things That Matter Most

**First, the HBM franchise is the investment.** Strip away HBM and SK Hynix is a competent but unremarkable second-place memory producer trapped in the same commodity cycle that has destroyed shareholder value for decades. HBM transforms the thesis: it gives SK Hynix pricing power, margin visibility, and strategic relevance that commodity DRAM and NAND never provided. The company's first-mover advantage — shipping the first HBM product in 2013, a decade before the market exploded — reflects genuine engineering foresight, not luck. But every aspect of the bull case ultimately rests on HBM demand durability and competitive defensibility. If Samsung closes the yield gap or NVIDIA diversifies supply aggressively, the premium that the market assigns to SK Hynix evaporates.

**Second, this is still a deeply cyclical business wearing a growth stock costume.** The financial analysis reveals a company that swung from negative 20% operating margins in 2023 to 40% operating margins in late 2024 — a 60-percentage-point swing in twelve months. No amount of HBM differentiation eliminates the fundamental physics of memory: high fixed costs, commodity pricing at the margin, and an industry structure where all three oligopolists face identical incentives to over-invest at the top. The Red Flags analysis is right that every memory upcycle has ended in overinvestment. The question is whether HBM's tighter customer relationships and longer qualification cycles introduce enough friction to moderate the amplitude of the next downturn. The honest answer is: partially, but not enough to call this a secular compounder.

**Third, the chaebol governance structure is an underappreciated discount factor.** The management analysis paints a picture of improving execution and disciplined capital allocation, but this sits within SK Group's cross-shareholding web where minority shareholder interests are structurally subordinate. The Solidigm acquisition — $9 billion for a struggling NAND business acquired partly because it served SK Group's strategic ambition to become a "total semiconductor" player — is the clearest example of group-level priorities potentially overriding pure shareholder value maximization. International investors receive less disclosure than they would from Micron, and the holding-company layering through SK Square adds opacity.

**Fourth, customer concentration is hiding in plain sight.** The symbiotic relationship with NVIDIA is presented as a strength, but the dependency runs both ways and tilts increasingly toward NVIDIA's bargaining power as Samsung and Micron qualify their HBM products. A world where NVIDIA has three viable HBM suppliers is a world where SK Hynix's margins compress materially.

## Key Tensions and Contradictions

The most striking tension in this analysis is between the Competitive Landscape section's conclusion that advantages are "widening" and the Red Flags section's warning that every cycle-peak advantage in memory has historically proved temporary. Both are correct, but they operate on different time horizons. The HBM moat is real today; the question is whether it persists for two years or ten. The Investment Checklist's split verdicts — YES on industry structure and competitive advantages, NO on recession resilience, PARTIALLY on almost everything else — accurately captures a company that is simultaneously best-in-class and structurally fragile.

## Open Questions for Further Research

1. **HBM contract structure:** What portion of HBM revenue is under firm contracts vs. flexible purchase orders? How far forward does real demand visibility extend?
2. **Solidigm valuation:** What is the realistic standalone value of Solidigm today, and how does it compare to the carrying value on SK Hynix's balance sheet? Is there impairment risk?
3. **Samsung's HBM trajectory:** What is Samsung's actual HBM3E/HBM4 yield progress? This is the single most important competitive variable.
4. **Through-cycle ROIC with HBM:** Can the company sustain positive ROIC through a downturn for the first time in its history, given the HBM mix shift?
5. **U.S. export control exposure:** What is the realistic scenario for the Dalian fab exemption, and what is the stranded asset risk?

## Quality Assessment

SK Hynix is a **high-quality cyclical** — the best-positioned player in a consolidated industry, riding a genuine structural demand wave. It is not a quality compounder. The HBM franchise justifies a premium to historical memory valuations but does not eliminate cyclical risk. An investor should own this stock when the cycle, valuation, and competitive position align favorably — and should be prepared to reduce exposure when peak margins, peak capex, and peak consensus optimism converge. The current moment, with all three likely near their zenith, warrants caution on entry point even as the long-term strategic position remains compelling.

## Priority Next Steps

1. Obtain and analyse the latest quarterly earnings (Q4 2025 / Q1 2026) for updated HBM revenue mix and margin data
2. Review Samsung's latest HBM yield and qualification disclosures
3. Build a through-cycle DCF model stress-testing HBM margin durability at various market share scenarios
4. Analyse SK Group's related-party transaction disclosures from the most recent annual report
5. Monitor U.S. semiconductor export control policy developments affecting the Dalian fab

# 8. Learning Resources

## Podcasts & Audio

- **Acquired — "TSMC" (October 2023)** — Ben Gilbert & David Rosenthal. While focused on TSMC, provides essential context on the semiconductor supply chain and foundry relationships that shape how memory companies like SK Hynix operate and compete.
- **Odd Lots (Bloomberg) — HBM and AI chip episodes** (2023–2025) — Tracy Alloway & Joe Weisenthal. Multiple episodes cover the AI hardware buildout, including deep dives on HBM demand dynamics and the memory supply chain bottleneck.
- **The Chip Letter** — Alan Patterson. A recurring series covering semiconductor industry news with regular attention to the memory sector, DRAM/NAND pricing cycles, and Korean chipmakers' strategic moves.
- **Moore's Law Is Dead — HBM coverage** (2023–2025) — Frequent deep dives on HBM3, HBM3E, and HBM4 technology roadmaps with technical detail on SK Hynix's lead over Samsung.

## YouTube & Video

- **Asianometry — SK Hynix and HBM videos** (Jon Y) — Arguably the single best video resource. Rigorous research on SK Hynix's history, HBM architecture, and memory industry economics.
- **Branch Education — "How does Computer Memory Work?"** — Exceptional 3D-animated explainers of DRAM cell architecture and memory chip manufacturing at the physical level.
- **TechTechPotato (Dr. Ian Cutress) — HBM and Memory Technology Videos** — Former AnandTech editor provides technically precise coverage of HBM specifications, packaging technology (TSV, hybrid bonding), and competitive dynamics.
- **Colossus — "Samsung & Memory Semiconductors" (Business Breakdown)** — Detailed breakdown of the memory business model, covering DRAM/NAND economics, cyclicality, and competitive positioning that applies directly to SK Hynix.

## Books

- **"Chip War: The Fight for the World's Most Critical Technology"** — Chris Miller (2022) — The essential starting point. Covers the Korean memory industry's rise, the Hynix bailout saga, and how memory became strategically vital.
- **"Memory: The Origins and Future of DRAM"** — Sungjoo Hong (2021) — Detailed technical and business history of the DRAM industry from its origins through modern HBM, with particular attention to Korean companies.
- **"The Everything Blueprint"** — James Ashton (2023) — Excellent context on how chip design and manufacturing ecosystems interact.

## Long-Form Articles & Research

- **Fabricated Knowledge (Substack)** — Doug O'Laughlin — One of the best independent semiconductor analysts. Essential pieces on memory industry structure, HBM economics, and SK Hynix's competitive moat vs. Samsung.

- **SemiAnalysis — Dylan Patel** — HBM and memory supply chain articles. Known for breaking stories on AI hardware supply chains, HBM allocation to NVIDIA, and advanced memory economics.
- **Chips and Cheese** — Explains why the DRAM industry consolidated to three players and how oligopoly dynamics drive boom-bust pricing cycles.
- **Nikkei Asia — Semiconductors** — Standout long-form features on SK Hynix's transformation from bailout recipient to HBM leader. Particularly strong on the Korean semiconductor beat.

## Earnings Calls & Primary Sources

- **SK Hynix Investor Relations** — Official source for quarterly earnings, annual reports, and investor presentations. Calls held in Korean with English transcripts.
- **Seeking Alpha — SK Hynix (OTC: HXSCF)** — English-language earnings call transcripts, typically available within 24 hours.
- **DART (Korean EDGAR)** — All official SK Hynix filings in Korean, including detailed financial statements and material disclosures.
- **SK Hynix Newsroom** — Surprisingly high-quality technical explainers on HBM, CXL, DDR5, and strategic announcements.
- **TrendForce** — Industry data for DRAM and NAND pricing, market share, and shipment volumes.

## Suggested Learning Path

Start with Chris Miller's *Chip War* for historical context, then watch Asianometry's SK Hynix and HBM videos for technical grounding, read Fabricated Knowledge for current industry analysis, and follow SK Hynix's earnings calls quarterly to track execution against the AI memory demand cycle.